AP & RAP Assignment 6

1. Prove: If $X_n \to X$ converges in probability, and $Y_n \to Y$ converges in probability, then $X_n \pm Y_n \to X \pm Y$ converges in probability, and $X_n Y_n \to XY$ converges in probability.

2. (a) Prove that if r.v.s $X_n \xrightarrow{\text{a.s.}} X$ and $X_n \xrightarrow{\mathcal{L}^1} Y$, then $X = Y$ a.s.

   (b) Construct r.v.s X, $X_n, n \geq 1$, such that $X_n \xrightarrow{\text{a.s.}} X$ but $X_n \nrightarrow X$ in $\mathcal{L}^p$ for any $p > 0$.

   (c) Construct r.v.s X, $X_n, n \geq 1$, such that$X_n \xrightarrow{\mathcal{L}^p} X$ for all $p > 0$ but $X_n \nrightarrow X$ a.s.

3. Prove:

   (a) Let $0 \leq X_n \leq X$ and $X \in \mathcal{L}^1$ If X_n converges to X in probability, then $X_n \to X$ converges in $\mathcal{L}^1$.

   (b) Let $X_1, X_2, \cdots \in \mathcal{L}^1$ with $\inf_n E(X_n) > -\infty$ and $X_n \downarrow X$ almost everywhere. Then, $X_n \to X$ converges in $\mathcal{L}^1$.

4. If $\{X_n, n \geq 1\}$ are r.v.s with $\sup_{n\geq 1} \mathrm{E}\,|X_n|^\beta < \infty$ for some $\beta > 0$, then $\{|X_n|^\alpha, n \geq 1\}$ is uniformly integrable (namely, uniformly p-integrable for $p = 1$) for $0 < \alpha < \beta$.